

FACTSHEET FS06 · Same-month seasonality

Do stocks repeat their best calendar month?

Henrik Gjerner · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

STRATEGY	ORIGIN	FAMILY
Return seasonality (cross-sectional)	Heston & Sadka (2008)	Calendar & seasonality
UNIVERSES	BACKTEST	BOOKS
US · EU · UK · DK · SCANDI · World (point-in-time)	Feb 2013 – Aug 2026, monthly, net of costs	Long/short and long-only (stocks with the highest past returns in the coming calendar month)
REBALANCE	STATUS	
Monthly	Descriptive factsheet, not a registered trial	

The claim. A stock that did well relative to others in, say, March tends to do well again in March in later years, at annual lags up to 20 years. Heston & Sadka (2008) found it in US stocks 1965–2002; Keloharju, Linnainmaa & Nyberg (2016) found the same across markets and asset classes. **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned -12.7% to +3.1% a year and was positive in 1 of the six; no universe passes the 2.87 gate (significantly negative in EU). The long-only book (stocks with the highest past returns in the coming calendar month) had a higher Sharpe than the equal-weight universe in 1 of the six (UK), with alphas of -6.2% to +2.5%, none past the gate. **What goes wrong (§10):** the main drags are costs of about 3.9% a year. None of the pre-declared fixes passes the gate in both windows. **Classification (§11):** Calendar & seasonality; its returns sit closest to the momentum cluster of the signal library. **360° view (§12):** nearest neighbours Size, Momentum 12-1; after those neighbours and the market, nothing is left (alpha $|t| < 2$ everywhere): the signal is redundant.

1. Headline performance

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long- only CAGR	Long- only Sharpe	Long- only max DD	EW CAGR	EW Sharpe	Size- weighted CAGR
US	500 / 10	-8.3%	-2.81	-10.0% (-3.3)	0.12	+12.0%	0.66	-35%	+14.1%	0.95	+15.3%
EU	238 / 10	-10.9%	-3.60	-9.5% (-3.1)	-0.13	+5.9%	0.42	-34%	+10.8%	0.78	+10.1%
UK	238 / 10	+3.1%	0.64	+2.6% (0.5)	0.05	+13.5%	0.78	-35%	+9.8%	0.74	+8.5%
DK	19 / 3	-12.7%	-2.74	-12.5% (-2.8)	-0.01	+5.5%	0.39	-43%	+12.3%	0.82	+10.3%

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long-only CAGR	Long-only Sharpe	Long-only max DD	EW CAGR	EW Sharpe	Size-weighted CAGR
SCANDI	62 / 5	-7.3%	-2.13	-5.9% (-1.6)	-0.11	+10.2%	0.69	-39%	+12.5%	0.91	+10.7%
World	968 / 10	-2.9%	-1.17	-2.9% (-1.1)	-0.00	+12.3%	0.71	-40%	+11.9%	0.80	+10.8%

Monthly, local currency (World in USD). L/S = equal-weighted stocks with the highest past returns in the coming calendar month minus stocks with the lowest, net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

Same-month seasonality on six stock universes, 2013-2026



Growth of 1, log scale, local currency, net of costs. Henrik Gjering - Project 10 factsheets

2. Strategy description

A stock that did well relative to others in, say, March tends to do well again in March in later years, at annual lags up to 20 years. Heston & Sadka (2008) found it in US stocks 1965–2002; Keloharju, Linnainmaa & Nyberg (2016) found the same across markets and asset classes.

Why it might work. Recurring demand and information flows tied to the calendar (earnings seasons, dividend timing, fund flows); Keloharju et al. (2016) argue it reflects seasonal variation in risk premia.

Rule	Original (Heston & Sadka (2008))	This factsheet
Signal	Average return in the same calendar month over the past 1–20 years	Average return in the coming calendar month over all prior years available (1 year in 2013, 13 by 2026)

Rule	Original (Heston & Sadka (2008))	This factsheet
Portfolio	Decile L/S, one month	Extreme quantile L/S, one month
Weighting	Equal weighted	Equal weighted

3. Data load

Universe	Prices	Membership	Names eligible (median)	Currency
US	Sharadar SEP <code>closeadj</code> (total return), 805 tickers incl. delisted	Point-in-time: top 500 by market cap each month-end (Sharadar filings × price, A06 method)	~499	USD
EU	Project1 yfinance cache, Adj Close (TR)	Point-in-time: union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced	~238	local (EUR, SEK, DKK, PLN)
UK	Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×)	Point-in-time: FTSE 100 + FTSE 250, quarterly snapshots; 75% priced	~242	GBP
DK	Project1 cache, Adj Close	Point-in-time: OMX Copenhagen 25, quarterly snapshots; 91% priced	~19	DKK
SCANDI	Project1 cache, Adj Close	Point-in-time: OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced	~62	local (DKK, SEK, EUR)
World	US Sharadar + UK + EU above	Point-in-time union of the three; local-currency returns summed, not converted	~963	mixed

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as `run_wiki_pit.py` / `run_eu_pit.py`), with a short list of verified ticker renames (e.g. `NZYM-B` → `NSIS-B`, `WDH` → `DEMANT`, `DAI` → `MBG`). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

- **Data gap:** Price history starts in 2011–2012, so the signal averages 1 prior year in 2013 and at most 13 by 2026; Heston & Sadka use up to 20. The early years test a much noisier signal than the paper's.

4. Signal creation

$SEAS_{i,t}$ = mean of $r_{i,m}$ over the months m with the same calendar month as $t+1$ in earlier years. Signals use only information up to month-end t and are traded in month $t+1$. Eligibility: in the point-in-time universe on any of the last five trading days of month t .

5. Model build

- Sort eligible stocks into equal-count groups (deciles ≥ 100 names, quintiles 50–99, terciles < 50 ; fixed per universe).
- **Long/short** = stocks with the highest past returns in the coming calendar month minus stocks with the lowest; **long-only** = stocks with the highest past returns in the coming calendar month; benchmarks: equal-weight universe and size-weighted universe.
- Equal weights within each leg, rebalanced monthly.
- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

6. Performance in detail

US (top 500, point-in-time), median 500 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-8.3%	-8.7%	12.5%	-0.67 [-1.14, -0.19]	-0.86	-72%	45%	-2.81	0.12	-10.0% (-3.3)	8.6%
Long-only (net)	+13.4%	+12.0%	20.4%	0.66 [0.22, 1.18]	0.93	-35%	63%	2.94	1.23	-4.3% (-2.2)	12.6%
Stocks with the highest past returns in the coming calendar month (gross)	+15.4%	+14.2%	20.4%	0.76 [0.31, 1.29]	1.13	-34%	64%	3.38	1.23	-2.3% (-1.2)	12.4%
Stocks with the lowest (gross)	+18.8%	+18.6%	18.3%	1.03 [0.57, 1.61]	1.70	-31%	66%	5.10	1.12	+2.7% (1.6)	10.6%
EW universe	+14.4%	+14.1%	15.1%	0.95 [0.52, 1.53]	1.52	-25%	68%	4.80	1.00	–	8.9%
Size-weighted universe	+15.4%	+15.3%	14.4%	1.07 [0.61, 1.64]	1.79	-24%	69%	5.27	0.90	+2.4% (1.8)	8.3%

EU (11 national blue-chip indices, point-in-time), median 238 names, 10 groups

DK (OMXC25, point-in-time), median 19 names, 3 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-12.7%	-13.0%	14.9%	-0.85 [-1.46, -0.27]	-1.04	-85%	37%	-2.74	-0.01	-12.5% (-2.8)	9.7%
Long-only (net)	+7.1%	+5.5%	18.3%	0.39 [-0.13, 0.96]	0.47	-43%	55%	1.42	1.03	-6.2% (-2.6)	10.8%
Stocks with the highest past returns in the coming calendar month (gross)	+8.6%	+7.2%	18.4%	0.47 [-0.05, 1.06]	0.62	-42%	56%	1.73	1.03	-4.6% (-2.0)	10.7%
Stocks with the lowest (gross)	+17.3%	+16.8%	18.4%	0.94 [0.39, 1.59]	1.55	-32%	62%	3.48	1.05	+3.9% (1.4)	10.1%
EW universe	+12.8%	+12.3%	15.6%	0.82 [0.30, 1.44]	1.26	-33%	64%	3.20	1.00	–	9.3%
Size-weighted universe	+11.2%	+10.3%	16.6%	0.68 [0.11, 1.32]	0.94	-40%	61%	2.52	0.91	-0.5% (-0.2)	10.8%

SCANDI (OMXC25 + OMXS30 + OMXH25, point-in-time), median 62 names, 5 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-7.3%	-7.7%	12.1%	-0.60 [-1.16, -0.04]	-0.80	-69%	41%	-2.13	-0.11	-5.9% (-1.6)	7.7%
Long-only (net)	+11.0%	+10.2%	15.8%	0.69 [0.14, 1.32]	1.06	-39%	62%	2.47	1.01	-2.0% (-0.9)	8.5%
Stocks with the highest past returns in the coming calendar month (gross)	+12.8%	+12.3%	15.8%	0.81 [0.25, 1.45]	1.32	-38%	62%	2.90	1.01	-0.1% (-0.0)	8.4%
Stocks with the lowest (gross)	+15.6%	+15.0%	17.6%	0.89 [0.32, 1.56]	1.35	-33%	65%	3.35	1.12	+1.2% (0.6)	10.8%
EW universe	+12.8%	+12.5%	14.1%	0.91 [0.38, 1.54]	1.42	-29%	67%	3.62	1.00	–	8.3%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Size-weighted universe	+11.2%	+10.7%	13.9%	0.80 [0.31, 1.37]	1.21	-27%	63%	3.26	0.95	-1.0% (-0.9)	8.6%

World (US + UK + EU, point-in-time), median 968 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-2.9%	-3.3%	9.6%	-0.30 [-0.79, 0.19]	-0.45	-48%	47%	-1.17	-0.00	-2.9% (-1.1)	6.3%
Long-only (net)	+13.4%	+12.3%	18.9%	0.71 [0.22, 1.27]	1.03	-40%	62%	2.95	1.14	-0.9% (-0.5)	11.4%
Stocks with the highest past returns in the coming calendar month (gross)	+15.4%	+14.6%	18.9%	0.82 [0.33, 1.39]	1.25	-38%	63%	3.39	1.14	+1.1% (0.6)	11.2%
Stocks with the lowest (gross)	+13.5%	+12.3%	18.8%	0.71 [0.26, 1.26]	1.05	-34%	63%	3.10	1.15	-0.9% (-0.7)	11.1%
EW universe	+12.5%	+11.9%	15.7%	0.80 [0.33, 1.37]	1.20	-29%	64%	3.53	1.00	–	9.4%
Size-weighted universe	+11.4%	+10.8%	15.0%	0.76 [0.29, 1.33]	1.14	-28%	63%	3.23	0.94	-0.4% (-0.4)	9.0%

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

Year	US	EU	UK	DK	SCANDI	World
2013	-12.6%	-15.0%	+14.9%	-28.0%	-24.9%	-11.9%
2014	-12.5%	-9.3%	+27.2%	+9.9%	+1.9%	-5.9%
2015	-2.6%	-0.7%	+31.3%	-17.2%	+0.6%	+0.6%
2016	-11.7%	-16.1%	-19.9%	-14.3%	-5.6%	-11.8%
2017	-20.3%	-8.5%	+4.8%	-24.6%	-11.3%	-10.8%
2018	-3.1%	-21.2%	-6.9%	-10.1%	-14.9%	+1.1%
2019	-12.7%	-1.8%	+0.9%	-6.1%	+10.3%	-3.4%
2020	+15.6%	-6.2%	+17.5%	+19.2%	+3.2%	+6.9%
2021	-2.7%	-20.5%	-0.7%	-30.5%	-18.4%	+0.5%

Year	US	EU	UK	DK	SCANDI	World
2022	-28.1%	-1.1%	-18.2%	-7.7%	-6.8%	-17.3%
2023	+12.8%	-4.3%	+2.3%	-19.0%	+6.6%	+15.6%
2024	+5.8%	-23.8%	-18.4%	-13.2%	-5.2%	+5.7%
2025	-20.6%	-12.6%	+6.3%	+7.2%	-17.9%	-6.9%
2026	-15.1%	-6.7%	+3.3%	-26.4%	-15.1%	-2.6%

Calendar-year returns, long-only (net)

Year	US	EU	UK	DK	SCANDI	World
2013	+21.8%	+21.9%	+34.2%	+24.2%	+14.7%	+21.3%
2014	+10.1%	+1.8%	+25.6%	+18.4%	+18.1%	+5.0%
2015	+0.3%	+2.8%	+23.1%	+13.3%	+13.1%	+2.8%
2016	+2.6%	+7.2%	+12.8%	+8.1%	+28.5%	+0.6%
2017	+10.3%	+12.5%	+39.6%	+3.4%	+12.8%	+25.8%
2018	-9.2%	-24.9%	-9.2%	-15.5%	-15.9%	-9.8%
2019	+25.4%	+24.2%	+22.9%	+14.2%	+43.6%	+27.5%
2020	+34.5%	+11.8%	+20.3%	+55.3%	+32.6%	+26.5%
2021	+24.8%	+0.4%	+13.1%	+4.2%	+9.0%	+17.1%
2022	-30.7%	-15.0%	-24.3%	-25.9%	-22.8%	-27.5%
2023	+36.2%	+17.8%	+11.5%	-4.6%	+7.8%	+33.5%
2024	+28.3%	+1.4%	-8.0%	-5.0%	-0.9%	+19.1%
2025	+10.5%	+17.4%	+13.7%	+16.4%	+8.3%	+19.7%
2026	+19.3%	+14.1%	+27.5%	-8.4%	+7.5%	+23.8%

7. Trading record

The rebalance log per universe is in `results/xs_FS06_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

Universe	Rebalances	Names long / short	Turnover long leg / month	Turnover short leg / month	L/S months positive	Worst L/S month	When
US	163	50 / 50	83%	87%	45%	-13.7%	2022-01
EU	163	24 / 23	86%	83%	44%	-12.9%	2024-10
UK	163	24 / 23	85%	84%	53%	-16.3%	2016-06

Universe	Rebalances	Names long / short	Turnover long leg / month	Turnover short leg / month	L/S months positive	Worst L/S month	When
DK	163	7 / 6	65%	71%	37%	-12.4%	2021-12
SCANDI	163	13 / 12	78%	80%	41%	-10.1%	2025-04
World	163	97 / 96	85%	85%	47%	-10.3%	2022-01

Current book (signal at 31 August 2026; first 8 names per leg)

Universe	Long: stocks with the highest past returns in the coming calendar month	Short: stocks with the lowest
US	SNDK, GEV, APP, CEG, HOOD, PLTR, RDDT, TKO	CVNA, KVUE, DXCM, VTRS, EFX, THC, XYZ, INCY
EU	ZEAL.CO, JSW.WA, HM-B.ST, NKT.CO, BOL.ST, FORTUM.HE, BKT.MC, ITX.MC	PUIG.MC, DSFIR.AS, DTG.DE, BMPS.MI, BAVA.CO, ANE.MC, ORSTED.CO, EDEN.PA
UK	CWR.L, ALFA.L, GDWN.L, TRST.L, HWG.L, JD.L, AVON.L, EWG.L	MTLN.L, THG.L, OCDO.L, AML.L, MTRO.L, OXIG.L, DOCS.L, BCG.L
DK	ZEAL.CO, NKT.CO, NDA-DK.CO, JYSK.CO, GMAB.CO, TRYG.CO, VWS.CO, DSV.CO	BAVA.CO, ORSTED.CO, GN.CO, DEMANT.CO, AMBU-B.CO, ISS.CO, COLO-B.CO, NOVO-B.CO
SCANDI	ZEAL.CO, HM-B.ST, NKT.CO, BOL.ST, FORTUM.HE, SWED-A.ST, NOKIA.HE, SKA-B.ST	BAVA.CO, ORSTED.CO, METSO.HE, QTCOM.HE, EQT.ST, WRT1V.HE, GN.CO, DEMANT.CO

8. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West t, 6 lags.

Long/short

Universe	Factor model	Alpha / yr	t(alpha)	R²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	-8.4%	-3.29	0.25	mkt +0.16 (+2.9), value -0.63 (-4.4), momentum -0.30 (-2.6)
EU	French Europe 5F + WML	-9.2%	-2.71	0.12	HML -0.38 (-2.4)
UK	JKP GBR 7 themes	+0.8%	0.16	0.20	quality +0.78 (+2.6), short_term_reversal -0.57 (-2.9)
DK	JKP DNK 7 themes	-7.5%	-1.94	0.08	none t ≥ 2
SCANDI	French Europe 5F + WML	-4.2%	-1.35	0.14	HML -0.41 (-2.5)
World	JKP World 7 themes	-0.5%	-0.21	0.27	mkt +0.14 (+3.0), value -0.69 (-4.9), momentum -0.28 (-2.9), low_risk +0.33 (+2.2)

Long-only

Universe	Factor model	Alpha / yr	t(alpha)	R ²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	+1.0%	0.54	0.90	mkt +1.08 (+26.8), size -0.45 (-3.5), low_risk -0.35 (-2.2)
EU	French Europe 5F + WML	+2.4%	0.79	0.68	Mkt-RF +0.81 (+12.5), CMA -0.48 (-2.6)
UK	JKP GBR 7 themes	+7.4%	2.30	0.71	mkt +0.70 (+11.7), low_risk -0.71 (-3.1), quality +0.80 (+3.2)
DK	JKP DNK 7 themes	+2.1%	0.68	0.64	mkt +0.63 (+8.3), low_risk -0.69 (-4.4)
SCANDI	French Europe 5F + WML	+7.7%	2.62	0.58	Mkt-RF +0.67 (+10.1), WML -0.19 (-2.5)
World	JKP World 7 themes	+7.0%	3.07	0.82	mkt +0.89 (+14.5)

9. Statistical verdict

- **Gate:** $|t| > 2.87$ (2 books × 6 universes, Bonferroni 0.05/12, two-sided).
- **L/S mean:** positive in 1 of the six; passes in 0 of 6.
- **L/S CAPM alpha:** -12.5% to +2.6%, $|t|$ past the gate in 2 of 6.
- **Long-only:** Sharpe above the equal-weight universe in 1 of the six; alpha -6.2% to +2.5%, passing in 0 of 6.
- **Probabilistic Sharpe** of the L/S, $P(\text{true Sharpe} > 0)$: US 0.01, EU 0.00, UK 0.79, DK 0.00, SCANDI 0.01, World 0.13.
- **Publication decay:** gross L/S -10.0% to +11.9% a year in 2013–19 against -7.3% to +5.2% in 2020–26.

10. What goes wrong, and how to fix it

Method. Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FS03–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FS03–FS12, gate $p < 0.0020$.

10.1 Diagnosis (design window 2013–2019)

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
US	-6.3%	-0.02	-0.2%	-6.1%	-4.1%	-3.0%	-3.3%	173%	2013-05 -7%, 2019-06 -7%, 2016-01 -6%
EU	-5.6%	-0.06	-0.7%	-4.8%	-4.0%	-3.1%	-2.5%	168%	2013-04 -10%, 2016-01 -10%, 2016-03 -9%
UK	+11.9%	0.11	+1.4%	+10.5%	-4.1%	+8.7%	+3.2%	169%	2016-06 -16%, 2016-09 -7%, 2018-08 -6%

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
DK	-10.0%	-0.16	-2.4%	-7.6%	-3.3%	-4.3%	-5.7%	137%	2016-12 -11%, 2013-07 -8%, 2013-09 -8%
SCANDI	-2.0%	-0.27	-4.2%	+2.2%	-3.8%	+1.6%	-3.6%	159%	2013-09 -9%, 2017-08 -7%, 2018-04 -7%
World	-1.2%	0.04	+0.5%	-1.6%	-4.1%	+0.2%	-1.3%	171%	2013-05 -6%, 2013-04 -6%, 2013-06 -5%

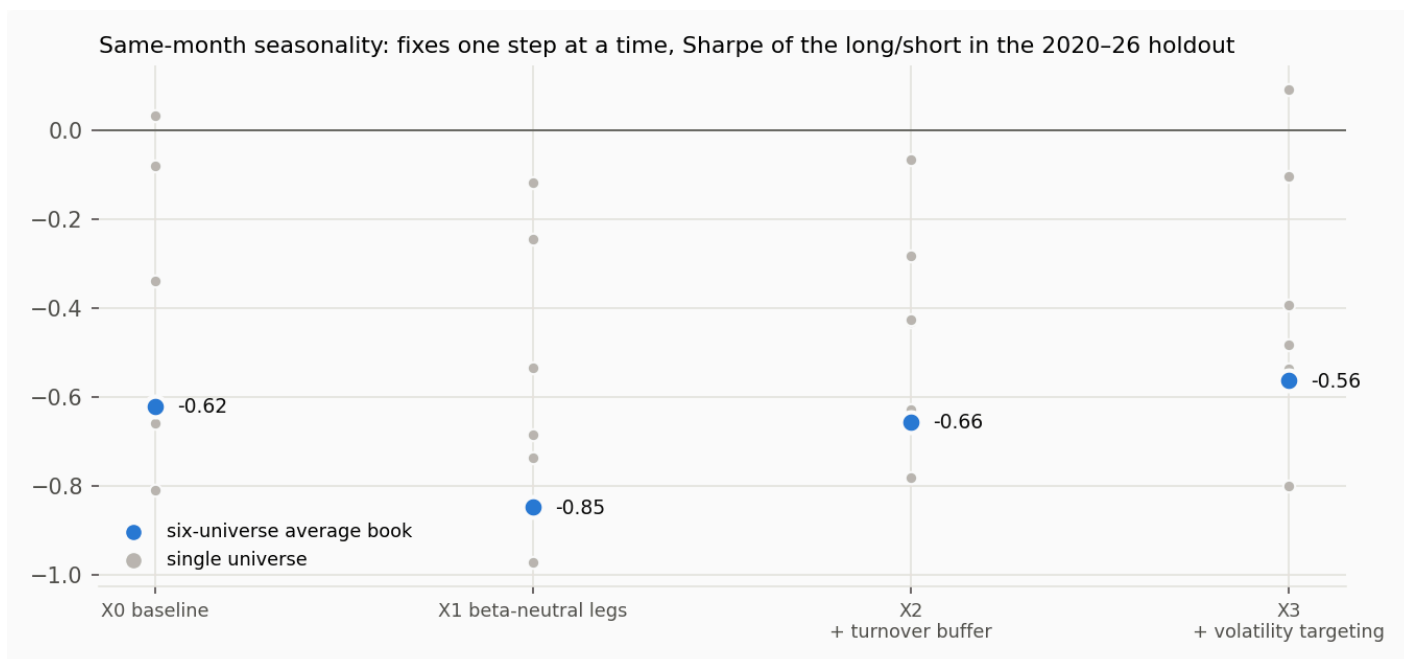
On average across the six universes the gross spread was -2.2% a year, of which the market exposure (beta -0.06) contributed -0.9%; the beta-adjusted spread (CAPM alpha) was -1.2%. Costs took 3.9% a year at 163% monthly turnover across both legs. The long leg beat the universe by +0.0% and the short leg lagged it by -2.2% a year, so most of the spread comes from the long side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages -1.8%, +1.1%, -0.4% a year.

10.2 The fix ladder

X0 baseline → X1 **beta-neutral legs** (each leg scaled by 1/its ex-ante beta) → X2 + **turnover buffer** (enter the extreme 1/q, keep a name until it leaves the extreme 3/q) → X3 + **volatility targeting** (scale the L/S to 10% a year using its trailing 6-month volatility, lagged; Barroso & Santa-Clara 2015; Moreira & Muir 2017).

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X0 baseline	-0.69	—	-0.62	—	—	-6.1%	-46%	-6.3% (-1.5)	3.9%
X1 beta-neutral legs	-0.72	-0.10 (p 0.848)	-0.85	-0.23 (p 0.916)	0 of 6	-7.7%	-51%	-7.8% (-2.0)	3.8%
X2 + turnover buffer	-0.71	+0.01 (p 0.453)	-0.66	+0.19 (p 0.030)	5 of 6	-5.5%	-45%	-5.4% (-1.4)	3.0%
X3 + volatility targeting	-0.62	+0.00 (p 0.496)	-0.56	+0.09 (p 0.170)	3 of 6	-5.0%	-45%	-4.8% (-1.1)	3.0%

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.

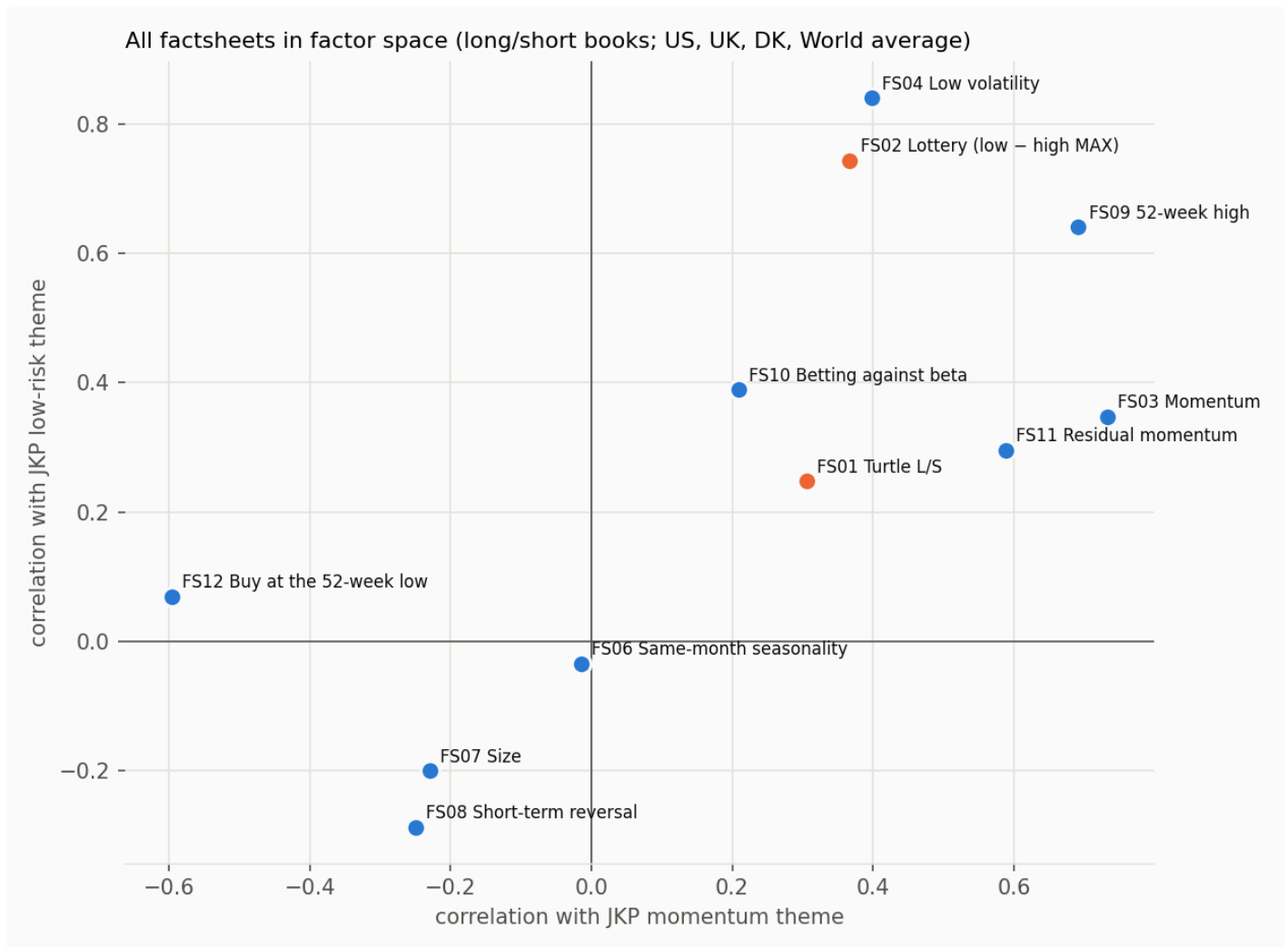


- **X1 beta-neutral legs:** -0.10 design, -0.23 holdout; hurts in both windows.
- **X2 + turnover buffer:** +0.01 design, +0.19 holdout; helps in both windows but does not pass the gate.
- **X3 + volatility targeting:** +0.00 design, +0.09 holdout; helps in both windows but does not pass the gate.

Where it ends: holdout Sharpe -0.62 → -0.56, return -6.1% → -5.0% a year (six-universe book).

11. Classification: where does it fit?

Dimension	Same-month seasonality
Series family	Calendar & seasonality
Academic style	Return seasonality (cross-sectional)
Signal	Price only (daily total returns)
Cross-section vs time series	Cross-sectional: stocks ranked against each other each month
Horizon and turnover	One month; 163% of the two legs replaced monthly
Market exposure	L/S beta -0.13 to 0.12
Payoff shape	Treynor–Mazuy γ -0.35 to 2.17 (t -0.4 to 1.9); worst 10% of market months -2.0% to +0.0% a month, best 10% -1.4% to +1.2%
Nearest factor theme	investment (4 of 6 universes), HML (2 of 6 universes)
Nearest library signals (returns)	Size (4 universes), 55-day breakout (3 universes), Last month return (3 universes)
Economic rationale	Recurring demand and information flows tied to the calendar (earnings seasons, dividend timing, fund flows); Keloharju et al. (2016) argue it reflects seasonal variation in risk premia.
Publication	Heston & Sadka 2008; Keloharju, Linnainmaa & Nyberg 2016
Where it fits	In the momentum cluster of the library; fully explained by its neighbours.



12. 360° view

The signal among the 19 price signals of the shared library (`code/library.py`), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **Momentum 12-1**.

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t $\geq +2 / \leq -2$
Same-month seasonality (this strategy)	seasonality	+1.00	+1.00	-1.8%	-8.5% to +7.2%	0 / 0
Last month return	short-term reversal	+0.02	+0.12	+0.9%	-3.2% to +8.4%	0 / 0
Momentum 12-1 (mirror)	momentum	+0.12	+0.11	+7.7%	+3.9% to +17.8%	5 / 0
55-day breakout	momentum	+0.01	+0.11	+0.7%	+5.6% to +12.6%	3 / 0
Net tail (MAX+MIN)	tail direction	+0.00	+0.08	-1.0%	-5.3% to +4.2%	0 / 0
Near 52-week high	momentum	+0.01	+0.08	+2.5%	+8.3% to +18.8%	6 / 0

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t $\geq +2$ / ≤ -2
MIN5	low risk	-0.04	+0.06	-2.0%	+2.4% to +13.2%	4 / 0
MIN (worst day)	low risk	-0.04	+0.05	-1.5%	+2.9% to +11.0%	3 / 0
Idiosyncratic vol	low risk	+0.03	-0.05	-1.5%	-10.4% to -0.8%	0 / 3
Residual momentum	momentum	+0.01	-0.04	+6.1%	+7.3% to +13.0%	5 / 0
Beta	low risk	+0.11	-0.04	+5.5%	-15.7% to -2.6%	0 / 3
Volatility	low risk	+0.06	-0.04	+3.0%	-10.3% to -3.7%	0 / 5
Range (MAX–MIN)	low risk	+0.05	-0.04	+1.2%	-11.3% to -2.3%	0 / 4
Volatility (252 days)	low risk	+0.09	-0.03	+5.1%	-10.3% to -0.2%	0 / 1
Size	size	+0.07	+0.02	-1.5%	-3.8% to +4.6%	0 / 0
Above 52-week low	momentum	+0.07	+0.02	+8.8%	+4.4% to +13.8%	4 / 0
Skewness	tail direction	+0.00	+0.02	-0.2%	-5.4% to +4.5%	0 / 1
MAX5	low risk	+0.06	-0.02	+2.6%	-8.5% to +0.4%	0 / 2
MAX (best day)	low risk	+0.05	-0.01	+1.9%	-9.3% to +1.0%	0 / 2

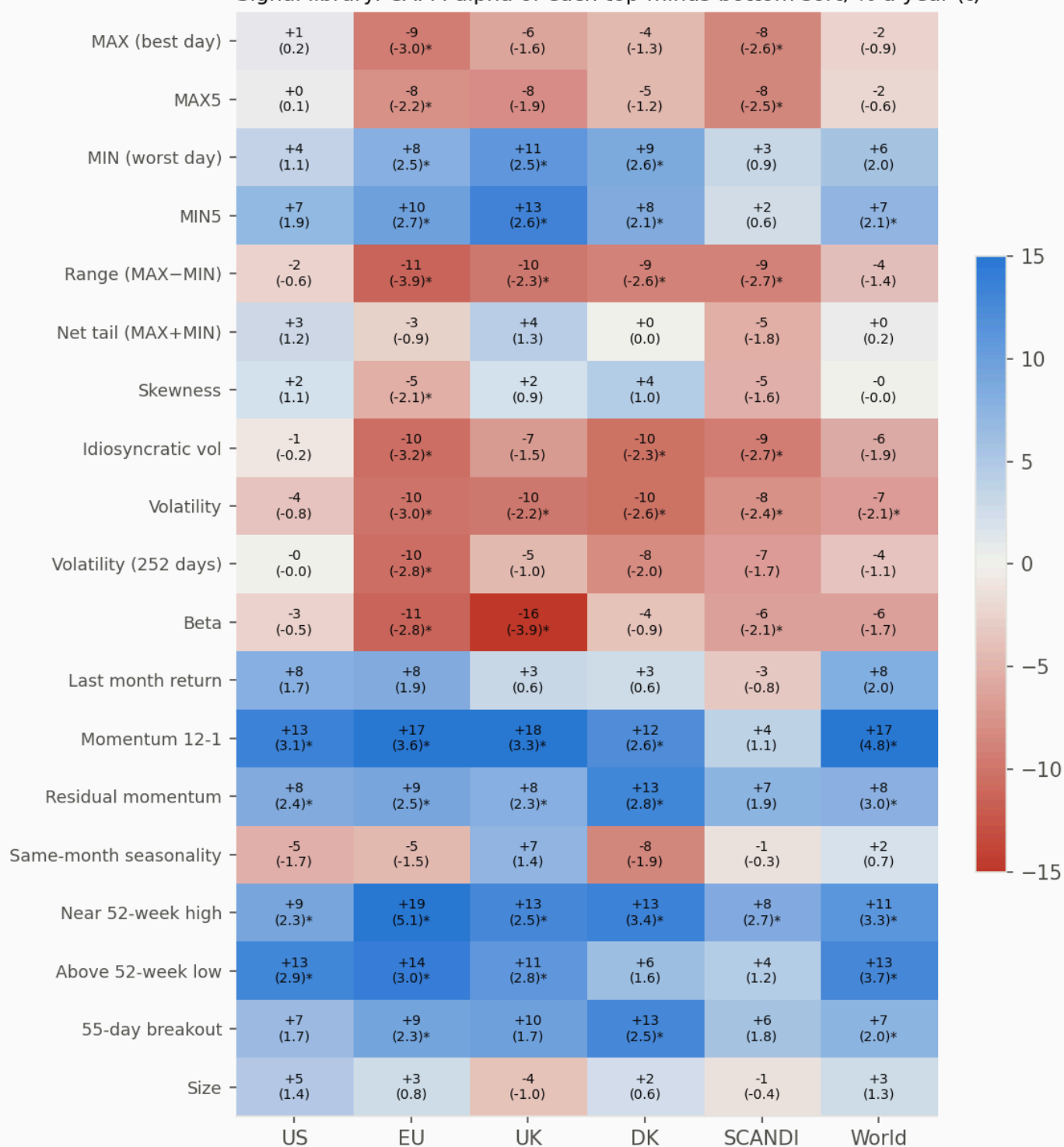
3×3 sort, next-month return a year
(US, EU, UK, SCANDI, World averaged)

low Same-month seasonality -	10.7%	12.4%	15.6%
middle -	10.5%	11.7%	13.5%
high Same-month seasonality -	10.0%	11.4%	14.8%
	low Momentum 12-1	middle	high Momentum 12-1

Holding Momentum 12-1 fixed (down a column), moving from low to high Same-month seasonality changes the return by -0.7, -1.0 and -0.8 points a year; holding Same-month seasonality fixed (along a row), moving from low to high Momentum 12-1 changes it by +4.9, +3.0 and +4.8. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): -1% vs -2% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages -1.8%, -1.8%, +1.1%, -0.4% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)



13. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.
3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.

4. **Data gap.** Price history starts in 2011–2012, so the signal averages 1 prior year in 2013 and at most 13 by 2026; Heston & Sadka use up to 20. The early years test a much noisier signal than the paper's.
5. **Not a registered trial.** Descriptive; the fix ladder is counted inside FS03–FS12 (25 trials).

14. Academic references

- Heston, S. & Sadka, R. (2008). Seasonality in the cross-section of stock returns. *Journal of Financial Economics*, 87(2), 418–445.
- Keloharju, M., Linnainmaa, J. & Nyberg, P. (2016). Return seasonalities. *Journal of Finance*, 71(4), 1557–1590.
- Newey, W. & West, K. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Politis, D. & Romano, J. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- Bailey, D. & López de Prado, M. (2012). The Sharpe ratio efficient frontier. *Journal of Risk*, 15(2), 3–44.
- Jensen, T. I., Kelly, B. & Pedersen, L. H. (2023). Is there a replication crisis in finance? *Journal of Finance*, 78(5), 2465–2518.
- Fama, E. & French, K. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
- McLean, R. D. & Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1), 5–32.
- Novy-Marx, R. & Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1), 104–147.
- Barroso, P. & Santa-Clara, P. (2015). Momentum has its moments. *Journal of Financial Economics*, 116(1), 111–120.
- Moreira, A. & Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance*, 72(4), 1611–1644.
- Treynor, J. & Mazuy, K. (1966). Can mutual funds outguess the market? *Harvard Business Review*, 44(4), 131–136.

15. Reproduce

`code/data.py` → `signals.py`, `signals2.py`, `library.py` → `xs.py` + `analyse_xs.py` FS06 → `build_xs.py`
FS06. Metrics use Project1 `InvestmentLibrary`. Licensed and cached prices are not included.